

When the School Bell Rings:
Analysis of Non-Motorist Crash Patterns Around Schools in Six
Maryland Jurisdictions During School Arrival and Dismissal Periods

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Abstract

School arrival and dismissal periods bring together students, families, staff, buses, private vehicles, pedestrians, bicyclists, and micromobility users within short time windows and concentrated school-access areas. This study examined whether school-related spatial and temporal contexts were associated with non-motorist crash patterns in selected Maryland jurisdictions. We constructed network-based school catchments, classified crashes by official school-calendar status, identified school-age involvement using expected school-age ranges, and compared crash frequency, reported crash share, and non-motorist injury severity across school-related contexts. The analysis was designed as a descriptive and bivariate association study. Overall, the results suggest that school geography and school timing are useful for identifying when and where non-motorist crash occurrence becomes more concentrated. School catchments, official school-open dates, and opening/dismissal windows were associated with higher non-motorist crash occurrence and a higher reported-crash share involving non-motorists. However, the evidence did not show that non-motorist injuries were significantly more severe in these school-related contexts. These findings support targeted school-area safety planning focused on arrival and dismissal periods, while also showing the need for future exposure-based and multivariable analysis.

Keywords: non-motorist safety; school catchments; school arrival and dismissal; reported-crash share; injury severity; Maryland

1. INTRODUCTION

School arrival and dismissal periods concentrate traffic from students, families, and staff. We studied whether the school-related spatial and temporal contexts were associated with higher non-motorist crash frequency, higher non-motorist crash likelihood, and greater non-motorist injury severity in six Maryland jurisdictions. We focused on 1-mile school catchment areas, official school-open dates, and school opening/dismissal windows because these contexts allow us to separate general proximity to schools from the specific days and hours when school travel is most likely to occur. We designed the analysis as a descriptive and bivariate association study rather than a causal or exposure-adjusted risk model.

2. LITERATURE REVIEW

Pedestrians, bicyclists, e-scooter riders, and other non-motorized road users are especially vulnerable in places where vehicle movements, pedestrian crossings, curbside activity, and turning conflicts are concentrated. School arrival and dismissal periods may create these conditions because they bring together students, parents, staff, buses, and neighborhood traffic within short time windows and small geographic areas. They may produce a distinct traffic-safety environment: more vehicles arrive near schools, more pedestrians and cyclists cross streets, and more conflicts occur at intersections, driveways, and curbside areas. This study examines whether school arrival and dismissal time windows are associated with increased frequency, likelihood, and severity of non-motorist crashes within school catchment areas in Maryland, specifically Prince George's County, Montgomery County, Howard County, Anne Arundel County, Baltimore City, and Baltimore County.

This research question fits within a broader shift in transportation planning toward centering safety issues. Across the United States, Vision Zero, Safe System, Complete Streets, and Safe Routes to School efforts all reflect the growing recognition that streets should be designed for the safety of all users, rather than primarily for motor-vehicle mobility (Federal Highway Administration, n.d.; National Complete Streets Coalition, n.d.; Vision Zero Network, n.d.). This concern is crucial near schools, where transportation planning affects family travel behavior and public health.

In Maryland and the Washington, DC-Baltimore region, this policy context is especially visible. Maryland has adopted a Vision Zero goal of eliminating vehicle-related deaths and serious injuries by 2030, highlighting pedestrian, bicyclist, school-access, and active-transportation safety needs through policies, such as the Safe Routes to School, Walk Maryland, and Complete Streets (Maryland Department of Transportation, 2019, 2024). Local and regional actors, such as Montgomery County, Prince George's County, Baltimore City, the Metropolitan Washington

Council of Governments, Washington Area Bicyclist Association, and other organizations have also emphasized pedestrian and bicyclist safety through Street Smart campaigns, local plans and reports, and related safety initiatives (Washington Area Bicyclist Association, 2024a, 2024b). Non-motorist safety, particularly near schools, is now a major planning concern in the study region. With that said, questions remain about how effective and comprehensive the current policies are.

This safety context has also changed as micromobility has become more common and diverse. E-scooters, e-bikes, shared bicycles, and other small mobility devices are making up a growing share of non-motorized travel in many urban and suburban areas (NACTO 2024). Earlier pedestrian and bicycle safety studies often relied on crash data from periods before these modes were widely used. Traditional motor-vehicle safety metrics do not always translate well to micromobility because risk estimates depend on how trips, exposure, crashes, injuries, and severity are defined (Karpinski et al., 2022). E-scooter crash and injury data may also be recorded inconsistently across police, hospital, and emergency-service datasets (Sexton et al., 2023). Therefore, focusing on the 2024 and 2025 crash records can help update earlier findings in light of the rapid growth of e-scooters and other modes.

Prior research shows that school environments are meaningful contexts for traffic-safety analysis. Clifton and Kreamer-Fults (2007) found that pedestrian-vehicle crashes near Baltimore City public schools were associated with certain school-site and neighborhood characteristics. This suggests that schools shape local travel behavior and crash risk rather than simply existing within a neutral traffic environment. Similarly, Yu and Zhu (2016) found that school siting together with the surrounding built environment influences motorist and pedestrian crashes; sidewalk coverage and local roads are associated with fewer crashes, while highways, commercial land uses, and transit stops are associated with more crashes. Also, non-motorist crash severity near schools may vary with socioeconomic context, which shows that school-zone risk is not evenly distributed across communities (Farid et al., 2025).

Additional school-zone research supports the need to consider both location and timing. When examining child pedestrian crashes near schools, Warsh et al. (2009) found that collision density was highest near schools and varied with school-related temporal patterns. School travel contributes meaningfully to morning traffic, and the long-term shift away from walking and biking toward private vehicle travel has increased the role of school trips in peak-period congestion (McDonald et al., 2011). Supporting this localized traffic peak, Kang et al. (2024) showed that school-run periods can measurably worsen traffic conditions around schools. These findings connect school catchments to both spatial exposure and time-specific congestion. A

crash near a school may reflect general neighborhood conditions, but a crash near a school during arrival or dismissal may be more directly connected to school-related travel.

By modeling pedestrian and bicyclist crash frequencies at Maryland intersections, Mahmoudi et al. (2023) found that crash patterns were associated with intersection design, signalization, traffic stress, pedestrian and bicycle volumes, vehicle volumes, transit conditions, land use, walkability, road network density, and some socioeconomic characteristics. Raihan et al. (2019) also concluded that bicyclist crash probabilities vary with road and intersection characteristics, such as lane width, speed limits, medians, sidewalks, signal control, bicycle activity, and bus stops. These support an intersection-level approach for assessing whether school-day activity poses risks beyond normal roadway and built-environment conditions.

Crash severity studies show that pedestrian injury outcomes are associated with age, crossing behavior, lighting, intersection proximity, vehicle type, speed, impairment, and built-environment characteristics (Clifton et al., 2009; Batouli et al., 2020). For bicyclists, vehicle speed, vehicle type, driver visibility, and roadway geometry were identified as injury severity factors (Robartes & Chen, 2017). Since the factors associated with whether a crash occurs may differ from those associated with how serious the resulting injuries are, our research treats crash frequency, crash likelihood, and crash severity as related but distinct outcomes.

Exposure remains a central challenge. For example, areas with more walking, biking, scooter use, and vehicle traffic may experience more crashes simply because more people are present. Cyclist crashes are often associated with bicycle kilometers traveled, vehicle kilometers traveled, land use, transit activity, road classifications, and bicycle facilities (Osama & Sayed, 2017). Even walkable and bikeable areas may experience higher pedestrian and cyclist crash risk, meaning that accessibility and safety are not the same (Osama et al., 2020). These built-environment conditions near schools can shape the risk of child pedestrian collisions (Rothman et al., 2014). Thus, there is a need to examine exposure effects vs. environmental risk effects when studying school catchments.

The literature also suggests that school-related traffic patterns may vary geographically. Liu et al. (2024) found that relationships between crashes and socioeconomic, built-environment, and transportation-network characteristics differ across space, while Amoh-Gyimah et al. (2016) showed the importance of accounting for unobserved spatial differences in pedestrian and bicycle crash models. For this Maryland study covering six urban and suburban counties, it is important to remember that school catchments in dense urban areas may operate differently from suburban or lower-density catchments, especially in terms of walking access, vehicle dependence, transit service, road design, and overall congestion.

Research Gap and Study Contribution

Previous studies establish that schools, intersections, roadway design, land use, exposure, age, socioeconomic conditions, and spatial context are related to pedestrian and bicyclist safety. However, the literature has not fully examined whether school opening and closing hours create distinct crash patterns within school catchments, including crash severity. School-area studies often focus on physical surroundings, while intersection-level crash models often focus on roadway characteristics and general exposure. Few studies combine school catchment boundaries, school calendars, school opening and closing times, crash timestamps, crash severity, and the ages of the people involved.

This study addresses that gap by separating geographic proximity from school calendar and time-of-day effects. In this study, we examined whether non-motorist crashes are more frequent and severe within school catchments, factoring in age, school open days, and the specific arrival and dismissal windows on school days. By incorporating 2024 and 2025 crash data, the study also updates earlier safety findings to reflect the current transportation environment, where e-scooters, e-bikes, and other micromobility modes are increasingly prevalent. The results can support targeted school-area safety interventions, such as intersection redesign, traffic calming, improved crosswalks, signal timing changes, curbside management, school bus routing, and improved drop-off and pick-up procedures.

3. RESEARCH DESIGN

The main objective of this study is to explore and test the association between non-motorist-involved crashes and the spatial and temporal context of elementary and secondary schools.. Therefore, we designed the study as a descriptive and bivariate association analysis rather than a causal or exposure-adjusted risk model because the available crash records did not include pedestrian volumes, bicyclist volumes, scooter volumes, school-trip volumes, or vehicle exposure measures.

To accomplish this, we first constructed network-based 1-mile school catchments and classified crashes by whether they occurred inside or outside those catchments. We then linked crash records to non-motorist and occupant records to identify school-age involvement, using the expected age ranges of nearby schools rather than assuming confirmed school enrollment. Next, we added official school-calendar status and school start/dismissal windows to distinguish general school proximity from the specific dates and time periods when school-related travel is most likely to occur. Finally, we evaluated the results in four outcome groups: descriptive school-age involvement, non-motorist crash frequency, reported-crash share or likelihood, and non-motorist injury severity.

The study, therefore, addresses four linked questions as summarized in Table 1. First, we ask what portion of crashes inside school catchments involved school-age persons. Second, we tested whether crashes involving non-motorists were more frequent in school-related contexts. Third, we test whether reported crashes in school-related contexts were more likely to involve non-motorists. Fourth, we test whether non-motorist injury severity was higher in school-related contexts. The first question is descriptive; the other three use bivariate statistical tests. Across all sections, we interpret statistical significance of our findings as evidence of association, not causation.

Table 1: Research questions and outcome-ordered hypothesis structure

Outcome order	Research focus	Hypothesis or expectation
Descriptive analysis	School-age involvement inside 1-mile school catchments	A measurable minority of inside-catchment crashes would involve at least one school-age person, but age alone would not confirm school-enrollment status or school-trip purpose.
Frequency	Non-motorist crash counts or rates across spatial, calendar, and school-time contexts	Non-motorist crash frequency would be higher inside school catchments, on official school-open dates, and during opening/dismissal windows.
Likelihood	Reported-crash share involving the relevant non-motorist outcome	Reported crashes in school-related contexts would be more likely to involve non-motorists.
Severity	Person-level ordinal non-motorist injury severity	Severity was tested empirically because the conditions associated with more crashes may differ from the conditions associated with more severe injuries.

Data Sources

As previously stated, we combined crash, person, school, street-network, and calendar data to connect each reported crash to its spatial location, temporal context, school-catchment status, age-based school context, and injury outcome. Four primary data sources were utilized:

- 1) The Maryland Approved Crash Reports provided access to the crash, vehicle, and person records of reported crashes.
- 2) Maryland educational facilities data supplied the school locations and school attributes
- 3) OpenStreetMap and OSMnx supplied the network basis for catchment construction
- 4) The Maryland State Department of Education's opening and closing calendars supplied the official school-open date classification.

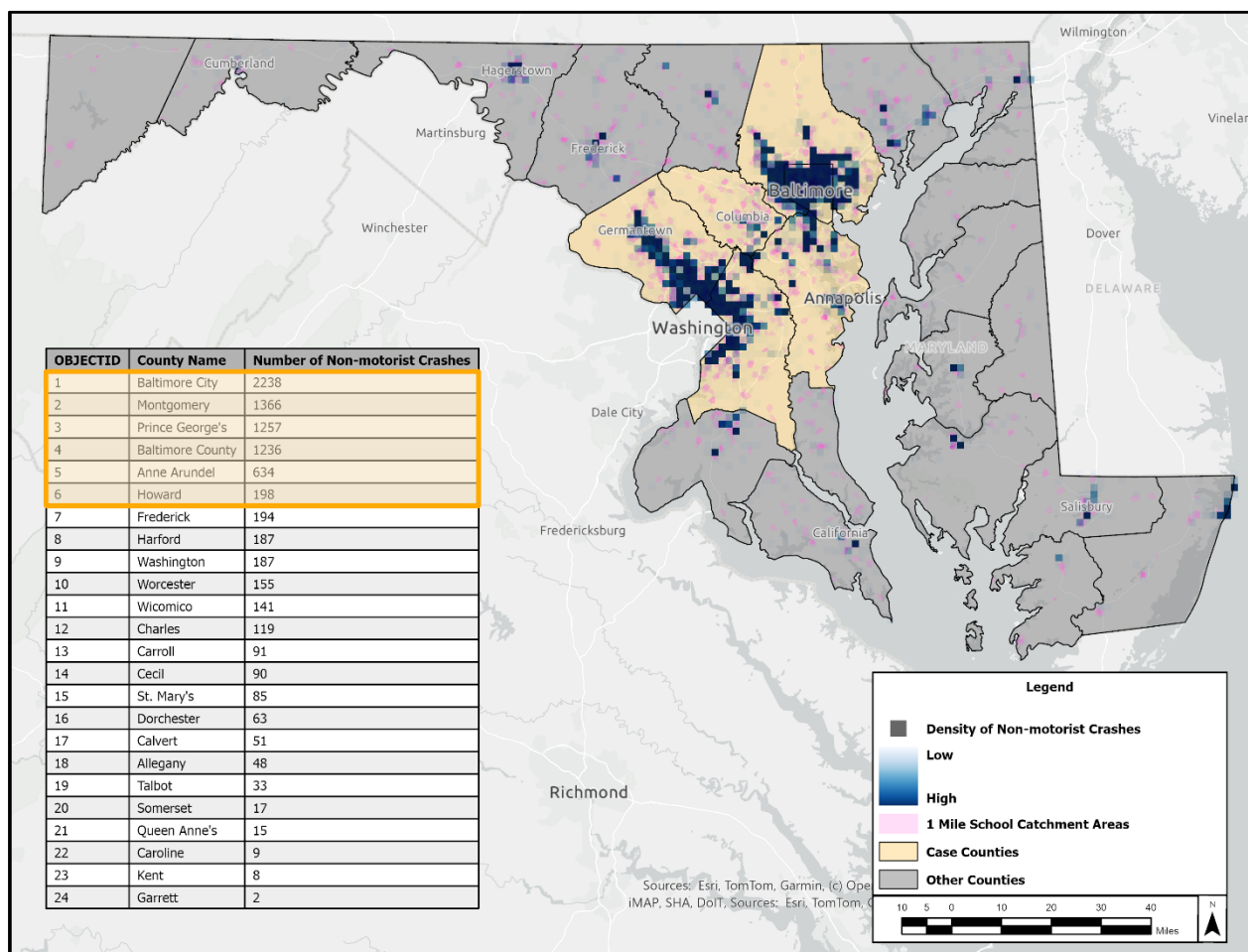
Study Area

The study area consists of six Maryland jurisdictions: Prince George's County, Montgomery County, Howard County, Anne Arundel County, Baltimore City, and Baltimore County. As evident in Figure 1, we selected these counties because:

- They had relatively high non-motorist crash frequencies in the 2024-2025 Maryland crash dataset.
- They capture urban, suburban, and lower-density contexts within Maryland.
- They form core parts of the Washington DC-Baltimore region and are connected by a multimodal transportation system.
- Together, they contain 941 elementary and secondary schools, representing almost 67% of Maryland educational facilities and a broad range of school types and age ranges.

Throughout the report, we will refer to these six jurisdictions as the study area.

Figure 1: Non-motorist involved crashes concentration in the study area



4. DATA PREPARATION AND METHODOLOGY

In this study, we used a staged descriptive and bivariate association design to evaluate how reported non-motorist crash patterns varied across school-related spatial and temporal contexts. Because the available data did not include pedestrian volumes, bicycle volumes, scooter volumes, school-trip counts, actual school attendance counts, vehicle volumes, or other exposure measures, we did not estimate a causal or exposure-adjusted crash-risk model. Instead, we compared crash frequency, reported-crash share, and injury severity across consistently defined school-related contexts.

School Catchment Classification

To classify crashes by school proximity, we used OpenStreetMap street-network data and the OSMnx package to construct 1-mile network-based school catchments. This approach was more appropriate than circular buffers because it represented proximity through the transportation network rather than straight-line distance. Each crash was classified as either inside or outside at least one 1-mile school catchment.

For crash (i), school-catchment status was defined as:

$$InsideCatchment_i = \begin{cases} 1, & \text{if crash } i \text{ occurred inside at least one study-area school catchment} \\ 0, & \text{otherwise} \end{cases}$$

This classification allowed us to compare reported crashes in school-access areas with crashes outside those areas. However, this measure should not be interpreted as a direct exposure measure because it does not include the number of pedestrians, bicyclists, scooter users, school travelers, or vehicles using those areas.

Handling Overlapping School Catchments

Some crashes fell within more than one school catchment. To avoid duplicating crashes or arbitrarily assigning them to only one school, we collapsed overlapping school matches into one crash-level school context. For crash (i), let (S_i) represent the set of schools whose catchments contained the crash. We defined the combined school-age range as:

$$MinAge_i = \min_{s \in S_i}(MinAge_s)$$

$$MaxAge_i = \max_{s \in S_i}(MaxAge_s)$$

Where ($MinAge_s$) and ($MaxAge_s$) are the expected minimum and maximum ages associated with school (s). A person (p) involved in crash (i) was classified as school-age when:

$$SchoolAge_{ip} = \begin{cases} 1, & \text{if } Age_{ip} \text{ is observed, } 0 \leq Age_{ip} \leq 100, \text{ and } MinAge_i \leq Age_{ip} \leq MaxAge_i \\ 0, & \text{otherwise} \end{cases}$$

Missing ages and ages outside the plausible 0–100 range were excluded from school-age classification. This method identified possible school-age involvement but did not confirm student enrollment, school-trip purpose, or connection to a specific nearby school.

For the opening/dismissal-window assignment, we used the same collapse logic for time of day. When a crash fell inside multiple school catchments, the crash-level school-time context used the earliest and latest relevant paired school start and dismissal times, rather than selecting only one school. This allowed overlapping school contexts to be represented without duplicating crash records.

Official School Calendar Classification

For calendar-sensitive analyses, we classified each school-system date as an official school-open date only when it met the documented calendar conditions. A date had to fall within the school system’s first and last school days, occur on a weekday, and not be listed as a weekend, local closure, public holiday, MSEA day, Thanksgiving break, winter break, spring break, other listed closing, or summer/non-session period.

For school system (k) on date (t), the official school-open indicator was defined as:

$$Open_{kt} = \begin{cases} 1, & \text{if } t \in [FirstDay_k, LastDay_k], Weekday_t = 1, \text{ and } t \notin Closures_k \\ 0, & \text{if } t \text{ was covered by the reference calendars but was not open} \\ Excluded, & \text{if } t \text{ was outside reference-calendar coverage} \end{cases}$$

The reference calendars covered the 2024–2025 and 2025–2026 school years. Therefore, dates in the January–August 2024 portion of the crash dataset were outside the official documented calendar coverage. We excluded those dates from calendar-sensitive analyses rather than replacing the official-calendar classification with a weekday-only assumption.

Frequency Measures

We used different frequency denominators for different research contexts because the appropriate unit of comparison changed across the analysis. For the spatial school-catchment comparison, the unit was the nearest-intersection/context group. For nearest-intersection/context group (j), the non-motorist crash count was:

$$NMCount_j = \sum_{i \in j} I(NM_i = 1)$$

where ($I(NM_i = 1)$) equals 1 when crash (i) involved at least one non-motorist and 0 otherwise. This aggregation avoided comparing only raw crash totals across inside- and outside-catchment areas, but it did not adjust for pedestrian, bicyclist, scooter, school-trip, or vehicle exposure.

For the official school-calendar comparison, the unit was the school-system date. For school system (k) on date (t), the inside-catchment non-motorist crash count was:

$$NMCount_{kt} = \sum_{i \in kt} I(NM_i = 1)$$

Zero-count school-system dates were retained because days with no non-motorist crashes are part of the calendar denominator.

For the opening/dismissal-window comparison, the unit was the crash-context 15-minute interval opportunity. Each eligible official school-day crash context was divided into 96 possible 15-minute intervals. The event rate for time-window group (g) was:

$$Rate_g = \frac{E_g}{T_g} \times 100,000$$

where (E_g) is the number of school-age non-motorist crash events in time-window group (g), and (T_g) is the number of eligible 15-minute interval opportunities in that same group. The rate was multiplied by 100,000 to make the interval-opportunity rate easier to interpret.

Reported-Crash Share Measure

We measured reported-crash share as the percentage of reported crashes in a given context that involved the relevant non-motorist outcome:

$$Share_g = \frac{NMCrashes_g}{TotalCrashes_g} \times 100$$

This measure is not a full exposure-adjusted risk measure. It compares the composition of reported crashes across school-related contexts. For the opening/dismissal-window analysis, the denominator was school-age-involved crash reports on official school-open dates, and the numerator was the number of those reports involving a school-age non-motorist.

Severity Measure

We measured non-motorist injury severity using person-level non-motorist injury status rather than the report-level crash severity field. Injury status was converted into an ordinal score:

$$SeverityScore_p = \begin{cases} 0, & \text{No apparent injury} \\ 1, & \text{Possible injury} \\ 2, & \text{Suspected minor injury} \\ 3, & \text{Suspected serious injury} \\ 4, & \text{Fatal injury} \end{cases}$$

For group (g), the mean non-motorist severity score was:

$$MeanSeverity_g = \frac{1}{N_g} \sum_{p=1}^{N_g} SeverityScore_p$$

where (N_g) is the number of non-motorist person records in group (g). Because the injury scale is ordinal, the mean severity score was interpreted as a descriptive summary rather than as proof that the distance between injury categories is equal. Group differences were evaluated with the Mann–Whitney U test.

Statistical Tests

We used Mann–Whitney U tests for skewed count distributions and ordinal severity comparisons. This test compares the rank distributions of two independent groups without assuming normally distributed outcomes. For group 1, the Mann–Whitney U statistic is:

$$U_1 = R_1 - \frac{n_1(n_1 + 1)}{2}$$

where (R_1) is the sum of ranks for group 1 and (n_1) is the number of observations in group 1.

We used chi-square tests for reported-crash share comparisons and interval-rate comparisons. The chi-square statistic is:

$$\chi^2 = \sum \frac{(O - E)^2}{E}$$

where (O) is the observed count and (E) is the expected count under independence. Across all analyses, statistically significant results are interpreted as evidence of association, not causation.

5. TESTING THE HYPOTHESES

The following sections test the study hypotheses in the order of descriptive school-age involvement, frequency of non-motorist-involved crash occurrence, reported-crash share involving non-motorists, and non-motorist injury severity. This outcome order separates descriptive composition from statistical tests of occurrence and injury outcomes. It also makes clear that a school-related context can be associated with higher crash frequency or a higher reported-crash share without necessarily being associated with more severe injuries.

Descriptive Analysis

In this section, we examined how school-age involvement and non-motorist person records were distributed across the six-county study area. These comparisons are descriptive and compositional. They describe the distribution of reported crashes and non-motorist person records in the dataset, not exposure-adjusted crash risk. For each group (g), the descriptive percentage was calculated as:

$$Percent_g = \frac{Count_g}{Denominator_g} \times 100$$

where ($Count_g$) is the number of crashes or person records in the group of interest, and ($Denominator_g$) is the relevant analysis denominator. The denominator changes by question: all inside-catchment crashes for general school-age involvement, inside-catchment non-motorist crashes for school-age involvement among non-motorist crashes, and non-motorist person records with valid age for the age-distribution comparison.

Table 2 shows that school-age involvement was a measurable but limited share of all crashes inside 1-mile school catchments. Among all 108,267 inside-catchment crashes, 9.02% involved at least one school-age person, 8.26% involved a school-age occupant, and 0.80% involved a school-age non-motorist. Among the 5,805 inside-catchment non-motorist crashes, however, 15.00% involved a school-age non-motorist and 17.24% involved any school-age person. This pattern supports the descriptive expectation that school-age involvement is not dominant among all inside-catchment crashes, but it becomes more visible when the analysis focuses specifically on non-motorist-involved crashes.

Table 2: School-age involvement among crashes inside 1-mile school catchments

Measure	Count	Denominator	Percent
Any school-age person involved	9,767	All inside-catchment crashes	9.02%
Any school-age occupant involved	8,939	All inside-catchment crashes	8.26%
Any school-age non-motorist involved	871	All inside-catchment crashes	0.80%
School-age non-motorist involved among inside-catchment non-motorist crashes	871	Inside-catchment non-motorist crashes	15.00%
Any school-age person involved among inside-catchment non-motorist crashes	1,001	Inside-catchment non-motorist crashes	17.24%

Table 3 compares school-age involvement across counties within the six-county study area. Baltimore City had the highest share of inside-catchment crashes involving school-age non-motorists, while Howard County had the highest share of inside-catchment crashes involving any school-age person. These county-level differences suggest that school-age involvement is not evenly distributed across the study area. However, these values should be interpreted as reported-crash composition rather than exposure-adjusted risk because the denominator does not include school travel volumes, walking volumes, bicycling volumes, scooter volumes, or vehicle volumes.

Table 3: School-age involvement among inside-catchment crashes by county/jurisdiction

County	Total Inside School Catchment	School-age any Person Count	School-age Non-motorist Count	School-age Occupant Count	School-age any Person Share	School-age Non-motorist Share	School-age Occupant Share
Baltimore City	28244	3091	384	2724	10.94%	1.36%	9.64%
Baltimore County	25058	1902	163	1748	7.59%	0.65%	6.98%
Prince George's	24328	1896	105	1794	7.79%	0.43%	7.37%
Montgomery	17906	1550	131	1427	8.66%	0.73%	7.97%
Anne Arundel	9818	961	66	900	9.79%	0.67%	9.17%
Howard	2913	367	22	346	12.60%	0.76%	11.88%

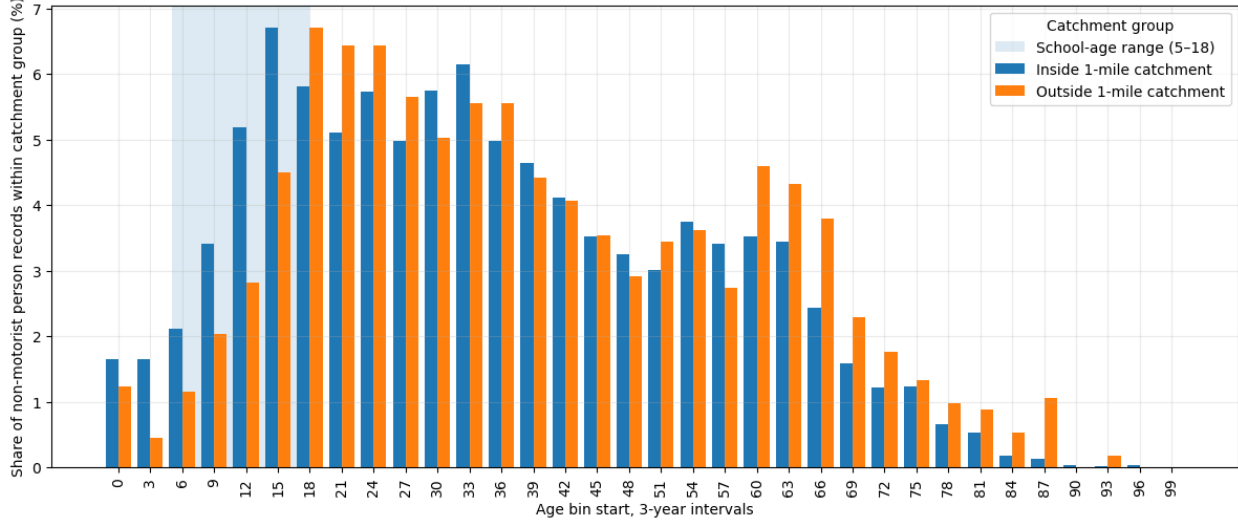
The age distribution used 3-year age bins to compare non-motorist person records inside and outside school catchments. For age bin (a) and catchment group (g), the plotted percentage was calculated as:

$$AgeShare_{ag} = \frac{NMPersons_{ag}}{NMPersons_g} \times 100$$

where ($NMPersons_{ag}$) is the number of non-motorist person records in age bin (a) and catchment group (g), and ($NMPersons_g$) is the total number of non-motorist person records in catchment group (g). This within-group percentage allowed the age composition of inside- and outside-catchment non-motorist records to be compared without being driven only by differences in raw group size.

Figure 1 shows that the inside-catchment age distribution was relatively elevated in several youth and young-adult bins, especially from late childhood through young adulthood, while both catchment groups tapered at older ages. The figure also shows that non-motorist involvement around school catchments is not limited only to school-age persons. Therefore, the age results support the need to distinguish between general non-motorist safety near schools and the narrower subset of possible school-age non-motorist involvement.

Figure 2: Distribution of non-motorist person ages in 3-year intervals by 1-mile school-catchment status. Percentages are calculated within each catchment group, so the figure compares age composition rather than raw person counts. The shaded range highlights the school-age interval used to interpret possible school-age involvement



Frequency of Non-motorist Involved Crash Occurrence

The frequency analyses tested whether non-motorist crash counts or event rates were higher in school-related contexts. We used three comparison units because each hypothesis required a different denominator: nearest-intersection/context groups for the spatial catchment comparison, school-system dates for the official-calendar comparison, and 15-minute interval opportunities for the opening/dismissal-window comparison.

For comparison unit (j), non-motorist crash frequency was calculated as:

$$NMCount_j = \sum_{i \in j} I(NM_i = 1)$$

where $(NMCount_j)$ is the number of non-motorist-involved crashes in comparison unit (j), and $(I(NM_i = 1))$ equals 1 when crash (i) involved at least one non-motorist. For the opening/dismissal-window analysis, the event rate was standardized by interval opportunities rather than by raw crash totals:

$$Rate_g = \frac{E_g}{T_g} \times 100,000$$

where (E_g) is the number of school-age non-motorist crash events in time-window group (g), and (T_g) is the number of eligible 15-minute interval opportunities in that same group. The effect ratio was calculated as the school-related value divided by the reference value. We used the Mann–Whitney U test for the spatial and calendar frequency comparisons because the count

distributions were skewed. We used a chi-square rate comparison for the interval-opportunity analysis because it compared event counts over defined time-interval opportunities.

Table 4 shows statistically significant frequency differences in all three contexts. Non-motorist crash frequency was higher inside 1-mile school catchments than outside catchments when standardized by nearest-intersection/context group. Inside catchments, official school-open dates had more non-motorist crashes per school-system date than non-opening dates. The largest frequency ratio appeared in the opening/dismissal-window analysis: school-age non-motorist crash events occurred at a substantially higher rate during opening/dismissal windows than during other official school-day times. This result provides strong evidence of association between school opening/dismissal windows and school-age non-motorist crash-event frequency, but it remains an association because the analysis does not include pedestrian, bicyclist, scooter, school-trip, or vehicle exposure volumes.

Table 4: Frequency tests for non-motorist crash occurrence across school-related contexts

Context	Comparison unit	School-related value	Reference value	Effect ratio	Test result
Spatial catchment	Nearest-intersection/context group	0.1284 non-motorist crashes	0.0595 non-motorist crashes	2.1595	Mann-Whitney U; $p < .001$
Official school calendar	School-system date	1.5996 inside-catchment non-motorist crashes	1.1256 inside-catchment non-motorist crashes	1.4211	Mann-Whitney U; $p < .001$
Opening/dismissal window	15-minute interval opportunity	39.0776 events per 100,000 intervals	6.2620 events per 100,000 intervals	6.2404	Chi-square = 471.3753, $df = 1$; $p < .001$

Reported-Crash Share Involving Non-Motorists

The reported-crash share analysis compared the percentage of reported crashes involving the relevant non-motorist outcome across school-related contexts. This measure is a composition measure, not a complete crash-risk measure, because it uses reported crashes as the denominator rather than pedestrian, bicyclist, scooter, school-trip, or vehicle exposure. For each context (g), the reported-crash share was calculated as:

$$Share_g = \frac{NMCrashes_g}{TotalCrashes_g} \times 100$$

For the opening/dismissal-window comparison, the denominator was school-age-involved crash reports on official school-open dates, and the numerator was the number of those reports involving a school-age non-motorist. We used chi-square tests of independence because each reported-crash share test compared a two-by-two contingency table of non-motorist involvement by context group:

$$\chi^2 = \sum \frac{(O - E)^2}{E}$$

where (χ^2) is the chi-square statistic, (O) is the observed count in a contingency-table cell, and (E) is the expected count under independence between the context group and non-motorist involvement.

Table 5 shows that reported crashes in school-related contexts were more likely to involve non-motorists as a share of reported crashes. The spatial contrast was the strongest proportional result: the non-motorist share was higher inside school catchments than outside them. On official school-open dates inside catchments, the non-motorist share was also higher than on non-opening dates. During opening/dismissal windows, school-age-involved crash reports were more likely to involve a school-age non-motorist than during other official school-day times. These results support the reported-crash share hypothesis, but they do not prove that school activity caused the difference because the denominator is reported crashes rather than population exposure or travel exposure.

Table 5: Reported-crash share tests for non-motorist involvement across school-related contexts

Context	Denominator	School-related share	Reference share	Effect ratio	Test result
Spatial catchment	All reported crashes	5.3617%	2.2359%	2.3981	Chi-square = 794.1295, df = 1; p < .001
Official school calendar	Inside-catchment crashes with official-calendar coverage	6.1162%	4.9405%	1.2380	Chi-square = 46.6963, df = 1; p < .001
Opening/dismissal window	School-age-involved crashes on official school-open dates	14.4225%	9.0629%	1.5914	Chi-square = 26.1513, df = 1; p < .001

Severity of Non-motorist Involved Crash Occurrence

The severity analysis used person-level non-motorist injury status rather than the report-level crash severity field. We converted the ordered injury categories to a 0–4 ordinal score: 0 for no apparent injury, 1 for possible injury, 2 for suspected minor injury, 3 for suspected serious injury, and 4 for fatal injury. For group (g), the mean severity score was calculated as:

$$MeanSeverity_g = \frac{1}{N_g} \sum_{p=1}^{N_g} SeverityScore_p$$

where (N_g) is the number of non-motorist person records in group (g). Because the injury scale is ordinal, the mean was reported as a descriptive summary rather than as proof of equal spacing

between injury categories. Mann–Whitney U tests were used because the test compares rank distributions across two independent groups without assuming normally distributed severity scores. For group 1, the test statistic is:

$$U_1 = R_1 - \frac{n_1(n_1 + 1)}{2}$$

where (R_1) is the sum of ranks in group 1 and (n_1) is the number of observations in group 1.

Table 6 does not show statistical evidence that non-motorist injury severity was higher in school-related contexts. In the spatial and calendar comparisons, the school-related mean severity scores were slightly lower than the reference scores, and neither difference was statistically significant. During opening/dismissal windows, the mean severity score and serious/fatal share were descriptively higher than during other official school-day times, but the Mann–Whitney U test was not statistically significant. Therefore, the severity results should be interpreted differently from the frequency and reported-crash share results. School-related contexts were associated with more non-motorist crash occurrence and a higher reported-crash share involving non-motorists, but they were not associated with a statistically significant increase in non-motorist injury severity.

Table 6: Non-motorist injury-severity tests across school-related contexts

Context	School-related mean	Reference mean	Serious/fatal share	Mean difference	Test result
Spatial catchment	1.7278	1.7579	17.0447% vs 18.9474%	-0.0301	Mann-Whitney U; p = .3498; not significant
Official school calendar	1.7031	1.7508	15.6977% vs 18.3183%	-0.0476	Mann-Whitney U; p = .1191; not significant
Opening/dismissal window	1.7595	1.6651	14.7679% vs 9.7674%	0.0944	Mann-Whitney U; p = .1498; not significant

6. CONCLUDING REMARKS

Across the research questions, we observed a consistent pattern for crash occurrence but not for injury severity. School catchments were associated with higher non-motorist crash frequency and a higher reported-crash share involving non-motorists. Official school-open dates were also associated with higher inside-catchment non-motorist crash frequency and reported-crash share. The strongest occurrence pattern appeared during opening/dismissal windows, when school-age non-motorist crash events were more concentrated than during other official school-day times.

The school-age results add important nuance. School-age involvement represented a limited share of all inside-catchment crashes, but it became more prominent within the subset of inside-catchment non-motorist crashes. This suggests that school-age non-motorist safety is most

visible when the analysis focuses on non-motorist crashes and school-day time windows rather than on all crashes inside school catchments.

Regarding crash severity, however, we did not find statistically significant evidence that non-motorist injuries were more severe inside school catchments, on official school-open dates, or during opening/dismissal windows. We therefore conclude that the strongest evidence from this study concerns crash occurrence and reported-crash share rather than injury severity. These findings support targeted school-area safety planning, especially at intersections and during arrival and dismissal periods. Potential interventions include traffic calming, improved crossings, signal-timing adjustments, curbside management, safer pick-up and drop-off procedures, and school travel coordination. However, the results should not be interpreted as causal evidence without additional exposure and control data.

7. LIMITATIONS AND NEXT STEPS

This study has a few limitations. First, we used a descriptive and bivariate association design, so the results identify associations rather than causal effects. Second, the crash data did not include pedestrian volumes, bicycle volumes, scooter volumes, school-trip counts, vehicle volumes, or other exposure measures. As a result, higher crash frequency or reported-crash share may reflect more people and vehicles being present rather than higher exposure-adjusted risk.

Third, we used age as a proxy for possible school-age involvement. The crash records identify the ages of non-motorists and occupants, but they do not indicate whether a person was enrolled in a nearby school, traveling to or from school, or connected to a specific school. Fourth, the 1-mile catchment and opening/dismissal windows are analytical definitions. They provide consistent comparison units, but they cannot fully capture actual school travel routes, household travel behavior, school-specific drop-off procedures, bus operations, after-school activities, or informal travel patterns.

Fifth, police crash records may undercount or inconsistently classify some non-motorist and micromobility events, especially lower-severity incidents that do not result in a formal crash report. Sixth, crashes may be spatially and temporally clustered around the same intersections, schools, corridors, or school-system dates, meaning that some observations may not be fully independent. Seventh, calendar-sensitive analyses were limited to dates covered by the available official school calendars, so uncovered dates were excluded rather than classified by a weekday-only assumption. Finally, the study used several bivariate tests, so individual p-values should be interpreted as part of the overall pattern rather than as isolated proof of causal relationships.

Future work should add exposure and contextual controls where possible. Useful next steps include incorporating pedestrian, bicycle, scooter, school-trip, and vehicle volume data; testing

alternative catchment sizes and time windows; modeling roadway, intersection, land-use, and socioeconomic variables; accounting for spatial clustering; and applying multivariable or quasi-experimental methods to evaluate whether school-generated travel contributes to crash occurrence after controlling for exposure and built-environment conditions. Additional field observations or school travel surveys could also help distinguish confirmed student travel from broader school-age involvement

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9. APPENDIX

The additional materials of this report are all accessible through the following GitHub Repository. To access it click [here](#) or scan the QR Code below.

